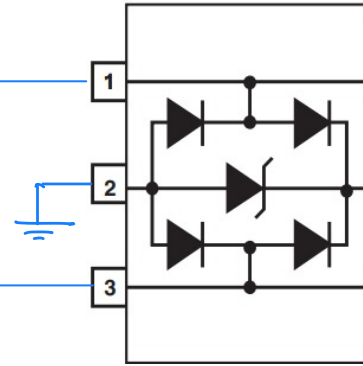
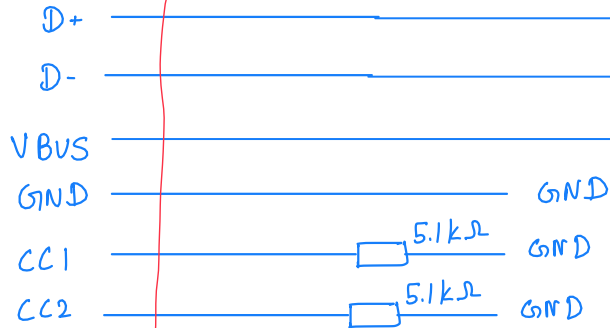


USB-C Female Jack

KOREAN HROPARTS
Elec Type-c-31-m-12



USBLC6-2SC6

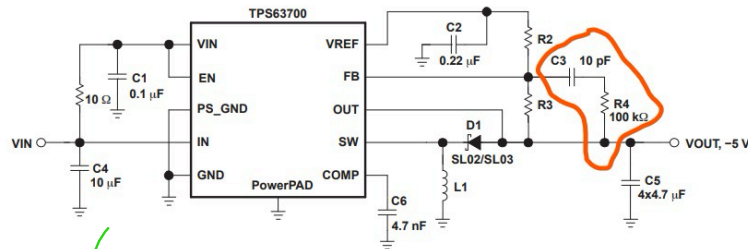
22Ω

ESP32
GPIO 20

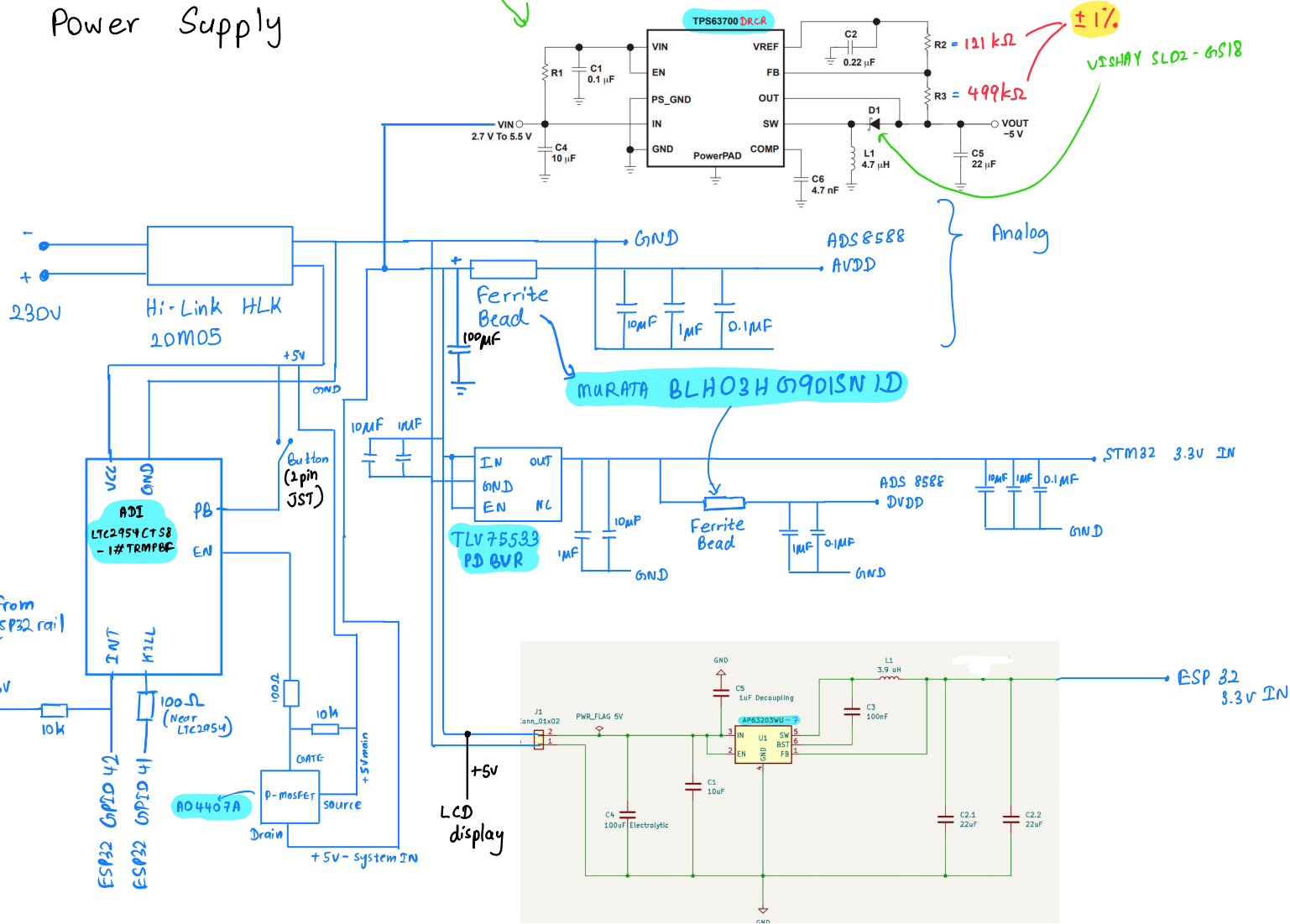
22Ω

ESP32
GPIO 19

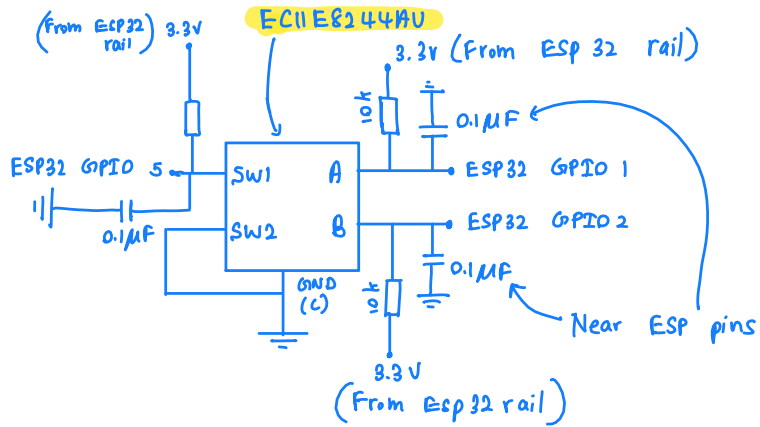
- Coupling Caps: Few mm distance between caps and Esp/LCD
No long traces
Thick ground paths
- SPI lines: Short, parallel } Continuous GND underneath the SPI Bus
Avoid 90° bends
- 5V Rail: 0.8 - 1.5 mm → Thickness
3.3V Rail: 0.8 - 1 mm → "
GND " : > 1 mm
- SPI lines: 0.25 mm
- UART T_x, R_x : 0.2 mm, Trace length < 10-15 cm
Continuous ground plane under traces
→ Don't route near; ADC analog inputs
High current paths



Power Supply

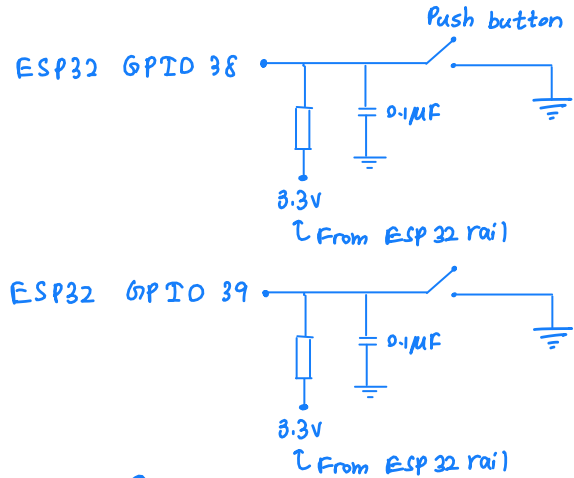


Rotary encoder and other buttons



Mark "DNP" near all 10k, 0.1µF caps and resistors of this rotary encoder on the pcb.

Back button / Enter button



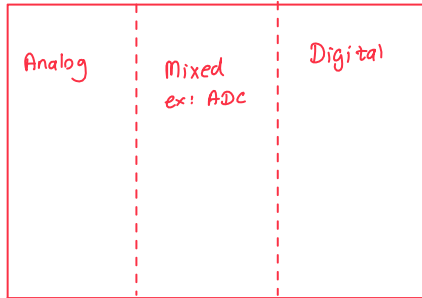
For this too

Use JST for rotary encoder and buttons.
(6-pin JST) (2-pin JST)

- 5V Rail : 0.8 - 1.5 mm $\rightarrow$ Thickness
- 3.3V Rail : 0.8 - 1 mm $\rightarrow$ "
- GND " : > 1 mm

Cap grounds : GND (No analog - digital separate grounding)

- Strong component partitioning over GND plane.



STM32 F411CEV6 - PIN 48

- 01) ADC - ADS588
- 02) ESP32 - UART connection
- 03) 3.3V Power supply

